CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

CO_2 emissions per delivery (ENV_FR_CO2)

DOMAIN

Transport	**Environment**	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Zero emission deliveries** *Improved environmental performance of urban freight distribution through a reduction in tank-to-wheel CO_2 emissions associated with each completed delivery.*	**ENV_FR**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Urban freight measures frequently aim to reduce the climate impact of delivery operations while maintaining the ability of logistics operators to serve urban customers. A reduction in CO_2 emissions per delivery indicates that each delivery is being performed with lower associated vehicle emissions, through a cleaner vehicle mix, reduced distance travelled per delivery, improved consolidation, higher delivery efficiency, or a shift towards locally zero-emission delivery modes.

This indicator measures the average tank-to-wheel CO_2 emissions associated with each completed freight delivery within a defined geographical area and reference period**. It is a relevant indicator when the policy action aims to reduce the emissions intensity of urban freight delivery activity. A successful action is reflected in a <u>LOWER</u> value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

The indicator assesses the average CO_2 emissions generated per completed delivery by urban freight delivery vehicles. It reflects whether delivery activity is becoming less carbon-intensive as a result of changes in vehicle technology, operating patterns, delivery consolidation, or the spatial efficiency of delivery tours. The indicator is calculated by multiplying the kilometres travelled by each delivery vehicle type by the corresponding theoretical tank-to-wheel CO_2 emission factor, summing the resulting emissions across all vehicle types, and dividing this total by the number of completed deliveries during the observation period.

The indicator is expressed as **$kgCO_2$/delivery**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **endogenously** in the supporting tool. The user fills in the kilometres travelled and deliveries completed per delivery vehicle type and the tool computes the indicator according to the method.

Method 1	
Survey delivery companies on kilometres travelled by delivery vehicles and deliveries made over a sample period	Significance: **0.75**

INPUTS

The following information is needed to compute the indicator:

- The **total kilometres travelled** by delivery vehicles in the pilot area during the chosen observation period, disaggregated by vehicle type.
- The **total number of completed deliveries** made by delivery vehicles in the pilot area during the chosen observation period, disaggregated by vehicle type.

The typical tank-to-wheel emission factors for selected vehicle types used by the tool in the calculation are:

- petrol vans: **72.5 gCO_2/km**;
- diesel vans: **79 gCO_2/km**;

- electric vans, hydrogen vans, cargo bikes and other alternative modes are considered locally zero-emission.

Source: EMEP/EEA air pollutant emission inventory guidebook 2023. https://www.eea.europa.eu/en/analysis/publications/emep-eea-guidebook-2023. Part B. 1.A.3.b.i-iv Road transport 2025. Tier 3-15.

EQUATIONS

The equation that is applied by the tool to quantify the indicator is:

$$\boldsymbol{CO_2DEL} = \frac{\sum_i KM_i \times EF_i}{\sum_i DEL_i}$$

Where:

$\boldsymbol{CO_2KM}$ = CO_2 emissions per kilometre travelled by delivery vehicles ($kgCO_2$/km)

KM_i = Total kilometres travelled by delivery vehicles of vehicle type i during the observation period (km)

EF_i = Theoretical tank-to-wheel CO_2 emission factor for vehicle type i ($kgCO_2$/km)

DEL_i = Total number of completed deliveries made by delivery vehicles of vehicle type i during the observation period

ALTERNATIVE INDICATORS

This indicator measures CO_2 emissions in relation to **delivery activity**. To assess CO_2 emissions in relation to vehicle movement, the CIVITAS Evaluation Framework also provides **ENV_FR_CO1: CO_2 emissions per km**, expressed in **$kgCO_2$/km**.